

Titration report



Method

Method ID	CDC_GranTED	Release state	Released
Name	EQP	Modified by	PredefinedUser
Modified on	2/3/2026 2:15:08 PM		

Summary

Samples

	Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.072 mL	OK
		Concentration R2	0.101 mol/L	OK
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.075 mL	OK
		Concentration R2	0.101 mol/L	OK
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.034 mL	OK
		Concentration R2	0.101 mol/L	OK
4	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.026 mL	OK
		Concentration R2	0.101 mol/L	OK
5	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.020 mL	OK
		Concentration R2	0.100 mol/L	OK
6	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.018 mL	OK
		Concentration R2	0.100 mol/L	OK

Statistics

R1

Mean	5.041 mL	Minimum	5.018 mL
Standard deviation	0.026 mL	Rel. standard deviation	0.514 %
Total samples	6	Samples excluded	0

R2

Mean	0.101 mol/L	Minimum	0.100 mol/L
Standard deviation	0.001 mol/L	Rel. standard deviation	0.513 %
Total samples	6	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
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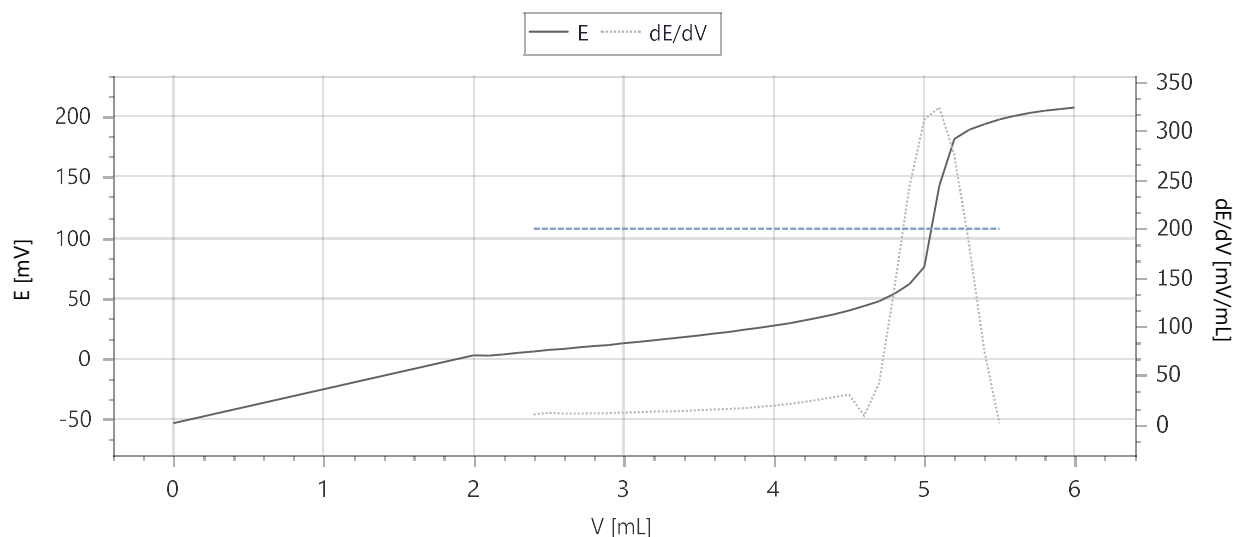
Titration report

LabX

0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.072 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.101 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.075 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.101 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.034 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.101 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.026 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.101 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.020 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.100 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Consumption	R1	5.018 mL	OK
0.1 M TRIS - 5 mL in 60 mL H ₂ O	Concentration	R2	0.100 mol/L	OK

Titration (EQP) [1] Scope 1/1, Sample 1/6

E - V & dE/dV - V Curve Scope 1/1, Sample 1/6



Measured Values Scope 1/1, Sample 1/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-53.3		25.0			
9	2.0000	2.7	2.0000	25.0	56.0		
43	2.1000	2.5	0.1000	25.0	-0.2		
47	2.2000	3.7	0.1000	25.0	1.2		

Titration report

LabX

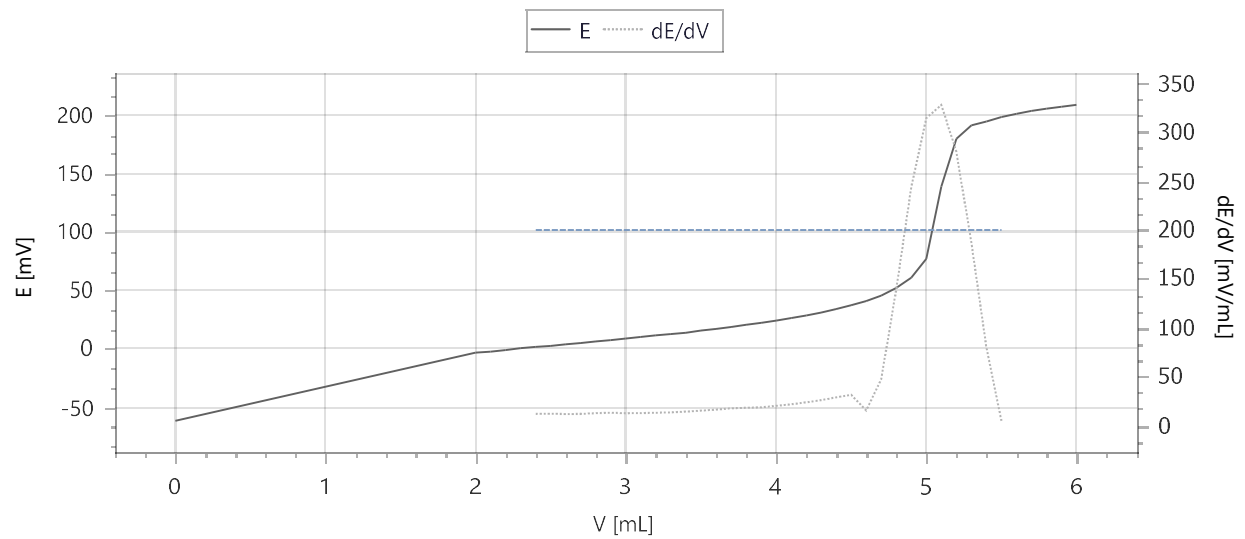
50	2.3000	5.0	0.1000	25.0	1.3	
54	2.4000	5.9	0.1000	25.0	0.9	9.93
57	2.5000	7.3	0.1000	25.0	1.4	11.71
60	2.6000	8.1	0.1000	25.0	0.8	10.78
63	2.7000	9.5	0.1000	25.0	1.4	10.81
66	2.8000	10.5	0.1000	25.0	1.0	11.16
72	2.9000	11.2	0.1000	25.0	0.7	11.28
76	3.0000	12.9	0.1000	25.0	1.7	11.88
79	3.1000	13.9	0.1000	25.0	1.0	12.14
82	3.2000	15.2	0.1000	25.0	1.3	12.97
86	3.3000	16.4	0.1000	25.0	1.2	13.15
90	3.4000	17.8	0.1000	25.0	1.4	13.46
94	3.5000	19.2	0.1000	25.0	1.4	14.35
97	3.6000	20.7	0.1000	25.0	1.5	14.98
100	3.7000	22.1	0.1000	25.0	1.4	15.63
104	3.8000	23.9	0.1000	25.0	1.8	16.35
108	3.9000	25.4	0.1000	25.0	1.5	17.67
111	4.0000	27.4	0.1000	25.0	2.0	18.99
115	4.1000	29.2	0.1000	25.0	1.8	20.60
120	4.2000	31.5	0.1000	25.0	2.3	22.76
123	4.3000	34.0	0.1000	25.0	2.5	25.04
127	4.4000	36.5	0.1000	25.0	2.5	27.74
132	4.5000	39.7	0.1000	25.0	3.2	30.08
137	4.6000	43.4	0.1000	25.0	3.7	8.57
140	4.7000	47.5	0.1000	25.0	4.1	42.46
146	4.8000	53.5	0.1000	25.0	6.0	138.34
151	4.9000	61.4	0.1000	25.0	7.9	242.13
158	5.0000	75.8	0.1000	25.0	14.4	311.61
	5.071757	123.8				324.90
173	5.1000	142.7	0.1000	25.0	66.9	324.07
177	5.2000	181.2	0.1000	25.0	38.5	275.64
188	5.3000	189.1	0.1000	25.0	7.9	181.81
196	5.4000	193.4	0.1000	25.0	4.3	74.78
199	5.5000	197.4	0.1000	25.0	4.0	0.86
202	5.6000	200.4	0.1000	25.0	3.0	
205	5.7000	202.6	0.1000	25.0	2.2	
208	5.8000	204.6	0.1000	25.0	2.0	
212	5.9000	205.9	0.1000	25.0	1.3	
216	6.0000	207.3	0.1000	25.0	1.4	

Titration (EQP) [1] Scope 1/1, Sample 2/6

E - V & dE/dV - V Curve Scope 1/1, Sample 2/6

Titration report

LabX



Measured Values Scope 1/1, Sample 2/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-61.4		25.0			
8	2.0000	-3.1	2.0000	25.0	58.3		
43	2.1000	-2.2	0.1000	25.0	0.9		
46	2.2000	-0.9	0.1000	25.0	1.3		
50	2.3000	0.7	0.1000	25.0	1.6		
53	2.4000	1.8	0.1000	25.0	1.1	11.81	
57	2.5000	2.7	0.1000	25.0	0.9	12.07	
61	2.6000	4.0	0.1000	25.0	1.3	11.63	
65	2.7000	5.3	0.1000	25.0	1.3	11.80	
68	2.8000	6.5	0.1000	25.0	1.2	12.57	
71	2.9000	7.7	0.1000	25.0	1.2	13.00	
75	3.0000	8.9	0.1000	25.0	1.2	12.45	
79	3.1000	10.3	0.1000	25.0	1.4	12.58	
82	3.2000	11.7	0.1000	25.0	1.4	12.99	
86	3.3000	12.9	0.1000	25.0	1.2	13.38	
90	3.4000	13.9	0.1000	25.0	1.0	14.20	
93	3.5000	15.8	0.1000	25.0	1.9	15.10	
97	3.6000	17.3	0.1000	25.0	1.5	16.20	
101	3.7000	18.9	0.1000	25.0	1.6	17.49	
105	3.8000	20.8	0.1000	25.0	1.9	18.10	
108	3.9000	22.5	0.1000	25.0	1.7	18.49	
112	4.0000	24.5	0.1000	25.0	2.0	20.07	
117	4.1000	26.7	0.1000	25.0	2.2	21.60	
120	4.2000	28.7	0.1000	25.0	2.0	23.65	
123	4.3000	31.3	0.1000	25.0	2.6	25.83	
127	4.4000	34.3	0.1000	25.0	3.0	28.74	
131	4.5000	37.4	0.1000	25.0	3.1	31.36	
134	4.6000	41.2	0.1000	25.0	3.8	15.29	
138	4.7000	45.7	0.1000	25.0	4.5	47.27	
143	4.8000	52.3	0.1000	25.0	6.6	139.08	
151	4.9000	61.1	0.1000	25.0	8.8	243.75	

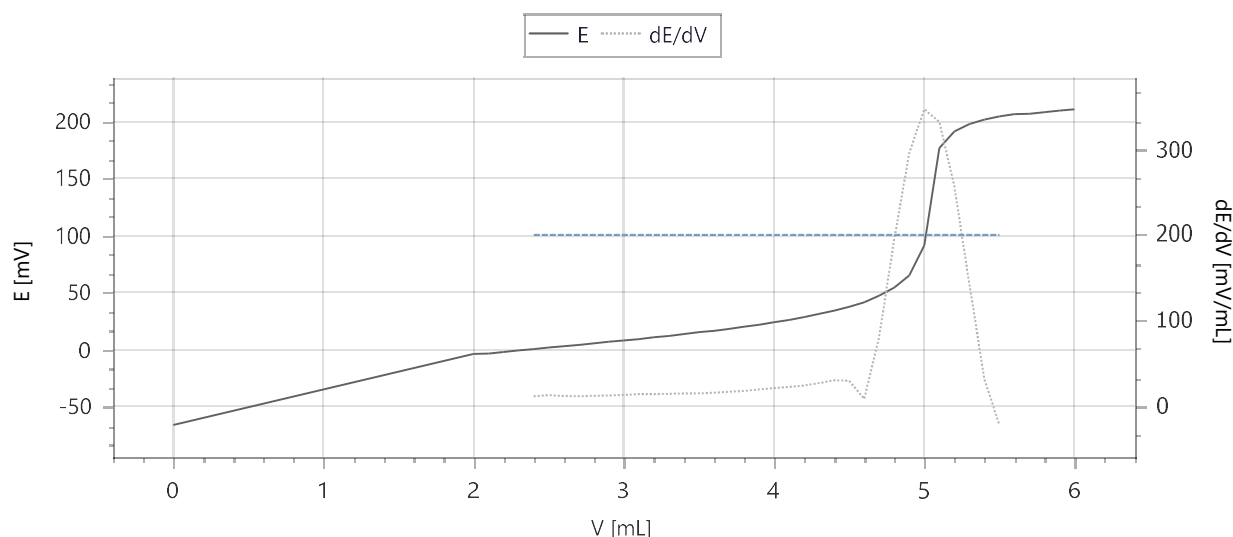
Titration report

LabX

160	5.0000	77.2	0.1000	25.0	16.1	314.33	EQP
	5.074687	123.5				328.68	
176	5.1000	139.2	0.1000	25.0	62.0	327.84	
180	5.2000	180.1	0.1000	25.0	40.9	280.14	
183	5.3000	191.6	0.1000	25.0	11.5	186.99	
193	5.4000	194.8	0.1000	25.0	3.2	79.98	
197	5.5000	198.6	0.1000	25.0	3.8	4.73	
200	5.6000	201.5	0.1000	25.0	2.9		
203	5.7000	203.9	0.1000	25.0	2.4		
206	5.8000	205.9	0.1000	25.0	2.0		
210	5.9000	207.5	0.1000	25.0	1.6		
213	6.0000	209.0	0.1000	25.0	1.5		

Titration (EQP) [1] Scope 1/1, Sample 3/6

E - V & dE/dV - V Curve Scope 1/1, Sample 3/6



Measured Values Scope 1/1, Sample 3/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-65.8		25.0			
9	2.0000	-3.5	2.0000	25.0	62.3		
42	2.1000	-3.4	0.1000	25.0	0.1		
46	2.2000	-1.9	0.1000	25.0	1.5		
49	2.3000	-0.6	0.1000	25.0	1.3		
52	2.4000	0.7	0.1000	25.0	1.3	11.52	
55	2.5000	2.1	0.1000	25.0	1.4	12.86	
59	2.6000	3.2	0.1000	25.0	1.1	11.90	
62	2.7000	4.1	0.1000	25.0	0.9	11.69	
65	2.8000	5.5	0.1000	25.0	1.4	12.06	
68	2.9000	6.9	0.1000	25.0	1.4	12.45	
71	3.0000	8.0	0.1000	25.0	1.1	13.17	
74	3.1000	9.2	0.1000	25.0	1.2	14.09	
78	3.2000	10.9	0.1000	25.0	1.7	14.07	
80	3.3000	12.1	0.1000	25.0	1.2	14.41	

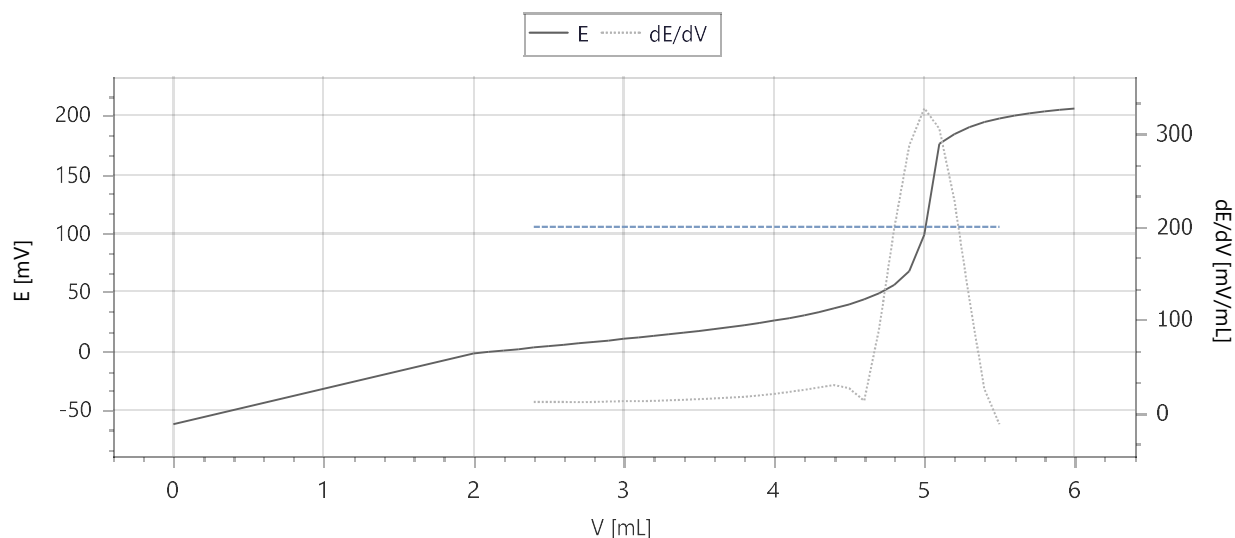
Titration report

LabX

84	3.4000	13.8	0.1000	25.0	1.7	14.84	
87	3.5000	15.4	0.1000	25.0	1.6	14.93	
90	3.6000	16.4	0.1000	25.0	1.0	15.70	
93	3.7000	18.3	0.1000	25.0	1.9	16.86	
96	3.8000	20.1	0.1000	25.0	1.8	17.70	
99	3.9000	21.9	0.1000	25.0	1.8	19.53	
104	4.0000	24.1	0.1000	25.0	2.2	21.07	
107	4.1000	26.0	0.1000	25.0	1.9	22.42	
111	4.2000	28.4	0.1000	25.0	2.4	24.16	
115	4.3000	31.3	0.1000	25.0	2.9	26.91	
119	4.4000	34.0	0.1000	25.0	2.7	30.25	
123	4.5000	37.5	0.1000	25.0	3.5	29.68	
126	4.6000	41.4	0.1000	25.0	3.9	8.36	
132	4.7000	47.2	0.1000	25.0	5.8	81.35	
138	4.8000	54.2	0.1000	25.0	7.0	195.10	
143	4.9000	65.0	0.1000	25.0	10.8	295.74	
148	5.0000	91.1	0.1000	25.0	26.1	346.91	
	5.034321	120.3				347.37	EQP
154	5.1000	176.3	0.1000	25.0	85.2	332.15	
158	5.2000	190.9	0.1000	25.0	14.6	257.01	
161	5.3000	197.3	0.1000	25.0	6.4	142.61	
164	5.4000	201.1	0.1000	25.0	3.8	31.87	
167	5.5000	203.9	0.1000	25.0	2.8	-22.07	
170	5.6000	205.8	0.1000	25.0	1.9		
178	5.7000	206.2	0.1000	25.0	0.4		
181	5.8000	207.6	0.1000	25.0	1.4		
184	5.9000	209.0	0.1000	25.0	1.4		
187	6.0000	210.2	0.1000	25.0	1.2		

Titration (EQP) [1] Scope 1/1, Sample 4/6

E - V & dE/dV - V Curve Scope 1/1, Sample 4/6



Measured Values Scope 1/1, Sample 4/6

Time	V	Meas. value	dV	Temp.	dE	dE/dV	EQP
04.02.2026 09:25:48				PredefinedUser			6/11

Titration report

LabX

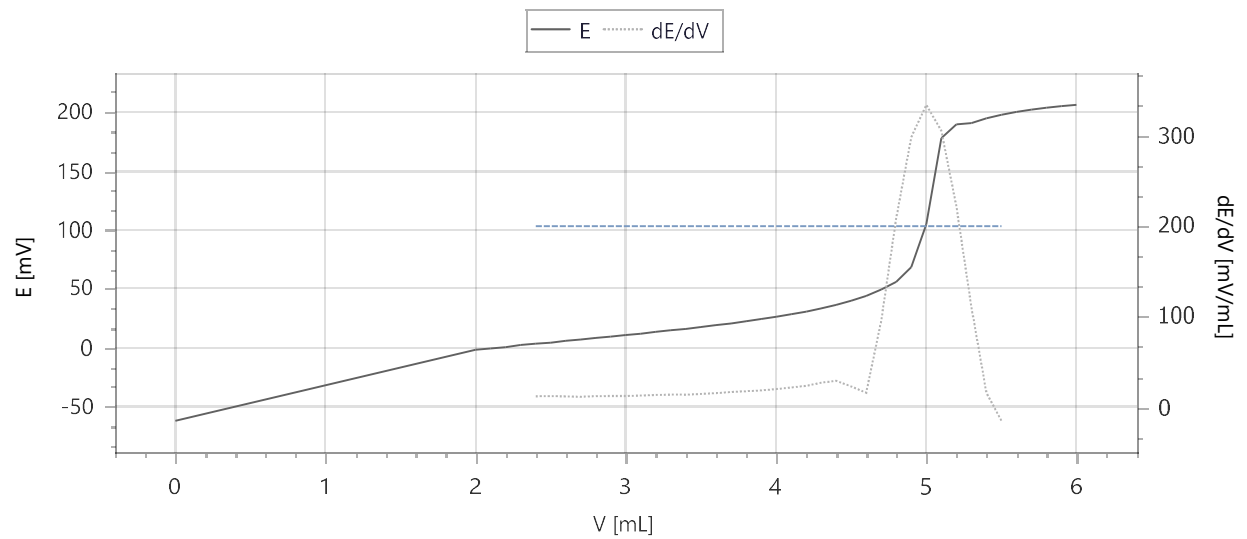
[s]	[mL]	[mV]	[mL]	[°C]	[mV]	[mV/mL]	
0	0.0000	-62.0		25.0			
10	2.0000	-2.0	2.0000	25.0	60.0		
44	2.1000	-0.6	0.1000	25.0	1.4		
48	2.2000	0.5	0.1000	25.0	1.1		
51	2.3000	1.7	0.1000	25.0	1.2		
54	2.4000	3.2	0.1000	25.0	1.5	12.08	
58	2.5000	4.5	0.1000	25.0	1.3	12.05	
62	2.6000	5.4	0.1000	25.0	0.9	12.05	
66	2.7000	6.7	0.1000	25.0	1.3	11.90	
69	2.8000	7.9	0.1000	25.0	1.2	12.06	
72	2.9000	9.1	0.1000	25.0	1.2	12.49	
76	3.0000	10.5	0.1000	25.0	1.4	12.87	
80	3.1000	11.7	0.1000	25.0	1.2	12.87	
82	3.2000	13.0	0.1000	25.0	1.3	13.23	
86	3.3000	14.3	0.1000	25.0	1.3	13.80	
89	3.4000	15.8	0.1000	25.0	1.5	14.43	
93	3.5000	17.2	0.1000	25.0	1.4	15.12	
97	3.6000	18.8	0.1000	25.0	1.6	15.84	
100	3.7000	20.5	0.1000	25.0	1.7	16.67	
103	3.8000	22.1	0.1000	25.0	1.6	17.58	
107	3.9000	23.9	0.1000	25.0	1.8	19.01	
111	4.0000	26.0	0.1000	25.0	2.1	20.73	
116	4.1000	28.1	0.1000	25.0	2.1	22.78	
119	4.2000	30.5	0.1000	25.0	2.4	25.11	
122	4.3000	33.3	0.1000	25.0	2.8	27.74	
126	4.4000	36.3	0.1000	25.0	3.0	30.18	
129	4.5000	39.7	0.1000	25.0	3.4	26.77	
133	4.6000	43.9	0.1000	25.0	4.2	13.42	
140	4.7000	49.2	0.1000	25.0	5.3	91.14	
145	4.8000	56.2	0.1000	25.0	7.0	198.38	
150	4.9000	67.9	0.1000	25.0	11.7	287.39	
157	5.0000	98.8	0.1000	25.0	30.9	327.18	
	5.025836	118.8				327.28	EQP
162	5.1000	176.1	0.1000	25.0	77.3	305.52	
177	5.2000	184.3	0.1000	25.0	8.2	228.84	
183	5.3000	190.3	0.1000	25.0	6.0	121.98	
186	5.4000	194.5	0.1000	25.0	4.2	25.96	
190	5.5000	197.6	0.1000	25.0	3.1	-11.90	
193	5.6000	199.9	0.1000	25.0	2.3		
197	5.7000	201.8	0.1000	25.0	1.9		
200	5.8000	203.5	0.1000	25.0	1.7		
203	5.9000	204.9	0.1000	25.0	1.4		
206	6.0000	206.1	0.1000	25.0	1.2		

Titration (EQP) [1] Scope 1/1, Sample 5/6

E - V & dE/dV - V Curve Scope 1/1, Sample 5/6

Titration report

LabX



Measured Values Scope 1/1, Sample 5/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-62.1		25.0			
11	2.0000	-1.8	2.0000	25.0	60.3		
45	2.1000	-0.6	0.1000	25.0	1.2		
48	2.2000	0.5	0.1000	25.0	1.1		
52	2.3000	2.3	0.1000	25.0	1.8		
54	2.4000	3.3	0.1000	25.0	1.0	12.26	
58	2.5000	4.3	0.1000	25.0	1.0	12.40	
61	2.6000	5.8	0.1000	25.0	1.5	12.17	
64	2.7000	7.0	0.1000	25.0	1.2	11.72	
67	2.8000	8.3	0.1000	25.0	1.3	12.40	
71	2.9000	9.2	0.1000	25.0	0.9	12.62	
75	3.0000	10.7	0.1000	25.0	1.5	12.55	
78	3.1000	11.8	0.1000	25.0	1.1	13.08	
81	3.2000	13.5	0.1000	25.0	1.7	13.79	
84	3.3000	14.7	0.1000	25.0	1.2	14.07	
89	3.4000	15.9	0.1000	25.0	1.2	14.05	
92	3.5000	17.6	0.1000	25.0	1.7	14.87	
95	3.6000	19.2	0.1000	25.0	1.6	15.74	
98	3.7000	20.5	0.1000	25.0	1.3	17.03	
102	3.8000	22.4	0.1000	25.0	1.9	17.85	
105	3.9000	24.4	0.1000	25.0	2.0	18.72	
109	4.0000	26.3	0.1000	25.0	1.9	20.21	
114	4.1000	28.3	0.1000	25.0	2.0	21.96	
118	4.2000	30.6	0.1000	25.0	2.3	23.80	
123	4.3000	33.2	0.1000	25.0	2.6	27.20	
127	4.4000	36.2	0.1000	25.0	3.0	29.38	
131	4.5000	39.7	0.1000	25.0	3.5	23.16	
136	4.6000	43.8	0.1000	25.0	4.1	16.20	
142	4.7000	49.2	0.1000	25.0	5.4	95.54	
146	4.8000	55.7	0.1000	25.0	6.5	208.88	
153	4.9000	68.4	0.1000	25.0	12.7	298.13	

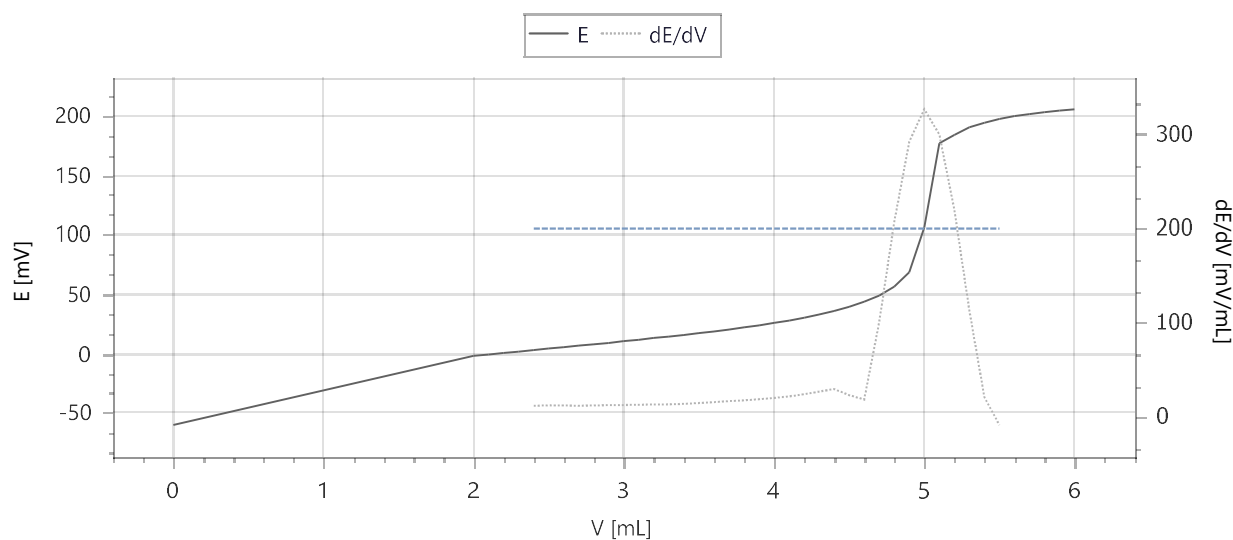
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160	5.0000	104.7	0.1000	25.0	36.3	333.69	EQP
	5.020034	119.3				333.69	
165	5.1000	177.6	0.1000	25.0	72.9	305.32	
168	5.2000	189.5	0.1000	25.0	11.9	222.10	
182	5.3000	190.6	0.1000	25.0	1.1	111.13	
186	5.4000	194.5	0.1000	25.0	3.9	16.38	
189	5.5000	197.6	0.1000	25.0	3.1	-14.78	
192	5.6000	200.0	0.1000	25.0	2.4		
195	5.7000	202.0	0.1000	25.0	2.0		
198	5.8000	203.4	0.1000	25.0	1.4		
202	5.9000	204.8	0.1000	25.0	1.4		
205	6.0000	205.9	0.1000	25.0	1.1		

Titration (EQP) [1] Scope 1/1, Sample 6/6

E - V & dE/dV - V Curve Scope 1/1, Sample 6/6



Measured Values Scope 1/1, Sample 6/6

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-59.9		25.0			
11	2.0000	-1.7	2.0000	25.0	58.2		
46	2.1000	-0.7	0.1000	25.0	1.0		
49	2.2000	0.5	0.1000	25.0	1.2		
53	2.3000	1.8	0.1000	25.0	1.3		
56	2.4000	3.0	0.1000	25.0	1.2	11.86	
60	2.5000	4.3	0.1000	25.0	1.3	12.40	
63	2.6000	5.5	0.1000	25.0	1.2	12.21	
67	2.7000	6.8	0.1000	25.0	1.3	11.92	
71	2.8000	7.9	0.1000	25.0	1.1	12.24	
75	2.9000	9.0	0.1000	25.0	1.1	12.51	
78	3.0000	10.5	0.1000	25.0	1.5	12.67	
82	3.1000	11.6	0.1000	25.0	1.1	12.99	
86	3.2000	13.2	0.1000	25.0	1.6	13.25	
89	3.3000	14.3	0.1000	25.0	1.1	13.40	

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94	3.4000	15.6	0.1000	25.0	1.3	13.93	EQP
98	3.5000	17.2	0.1000	25.0	1.6	14.78	
102	3.6000	18.6	0.1000	25.0	1.4	15.76	
106	3.7000	20.3	0.1000	25.0	1.7	16.85	
108	3.8000	22.1	0.1000	25.0	1.8	17.66	
113	3.9000	23.8	0.1000	25.0	1.7	18.86	
117	4.0000	25.9	0.1000	25.0	2.1	20.22	
121	4.1000	27.8	0.1000	25.0	1.9	21.82	
125	4.2000	30.3	0.1000	25.0	2.5	24.14	
130	4.3000	32.9	0.1000	25.0	2.6	26.99	
134	4.4000	35.7	0.1000	25.0	2.8	29.59	
138	4.5000	39.3	0.1000	25.0	3.6	23.19	
144	4.6000	43.5	0.1000	25.0	4.2	18.50	
150	4.7000	48.8	0.1000	25.0	5.3	100.74	
156	4.8000	56.1	0.1000	25.0	7.3	206.94	
165	4.9000	68.3	0.1000	25.0	12.2	292.16	
172	5.0000	105.9	0.1000	25.0	37.6	326.74	
	5.018051	118.6				326.75	
177	5.1000	176.5	0.1000	25.0	70.6	299.89	
192	5.2000	183.7	0.1000	25.0	7.2	219.93	
195	5.3000	190.0	0.1000	25.0	6.3	113.03	
198	5.4000	193.9	0.1000	25.0	3.9	21.11	
202	5.5000	197.1	0.1000	25.0	3.2	-8.58	
205	5.6000	199.4	0.1000	25.0	2.3		
209	5.7000	201.1	0.1000	25.0	1.7		
212	5.8000	202.7	0.1000	25.0	1.6		
215	5.9000	203.9	0.1000	25.0	1.2		
218	6.0000	205.2	0.1000	25.0	1.3		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Titration report

LabX

Responsible User

Time	Login name	Full name
03.02.2026 14:22:20	PredefinedUser	Administrator